

MANUFACTURING TECHNOLOGY

Manufacturing is the transformation of materials into products to meet human needs and wants. Manufacturing technology courses provide students with opportunities to acquire knowledge and skills in the areas of mechanical engineering, robotics and control systems, computer-aided manufacturing, industrial maintenance, precision machining, welding, and sheet metal.

Students will use a broad range of tools and equipment while acquiring engineering, fabrication, and problem-solving skills through the design and fabrication of various projects. Manufacturing courses will also help students understand the effects of manufacturing technology on the environment and society.

Manufacturing courses are designed for students heading to an apprenticeship, to college or university, or directly to the workplace in their pursuit of careers such as tradesperson, technician, technologist, engineer, or any of the skilled entry positions.

The list of approved emphasis areas for manufacturing technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Manufacturing Engineering Technology, Grade 11

University/College Preparation

TMJ3M

This course enables students to develop knowledge and skills related to design, process planning, control systems, and quality assurance. Students will use a broad range of tools and equipment and will combine modern manufacturing techniques and processes with computer-aided manufacturing as they develop critical decision-making, problem-solving, and project-management skills. Students will develop an awareness of environmental and societal issues related to manufacturing and will learn about pathways leading to careers in the industry.

Prerequisite: None

A. MANUFACTURING TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of design and other problem-solving processes used to develop solutions and products in response to challenges in manufacturing technology;
- A2.** demonstrate an understanding of the steps and processes required to produce a product;
- A3.** demonstrate an understanding of various types of control systems used in manufacturing;
- A4.** apply relevant mathematical skills, scientific concepts, and technological literacy and communication skills in developing a manufacturing system or process.

SPECIFIC EXPECTATIONS

A1. Design Process

By the end of this course, students will:

- A1.1** describe and explain the importance of the steps of a design process (*e.g., analysing the situation or context, developing design criteria, generating ideas, conducting research, communicating ideas, choosing the best solution, building a prototype, testing and modifying the prototype, undertaking a cost analysis*) or other problem-solving process used to solve a variety of manufacturing technology challenges (*e.g., use a problem-solving matrix for a robotics-related challenge arising from the production of the prototype*) (see pp. 21–23);
- A1.2** use technical drawings (*e.g., orthographic, pictorial, isometric, assembly*) as part of the design process to develop solutions to challenges or problems in manufacturing technology;
- A1.3** explain why technological concepts (*e.g., aesthetics, control, environmental sustainability/stewardship, ergonomics, fabrication, function, innovation, material, mechanism, power and energy, structure, safety, systems*) are important considerations in the design process (see pp. 7–8);
- A1.4** explain how the application of technological concepts in design or other problem-solving processes can result in products that better meet human needs or wants (*e.g., a ramp to replace a stairway, a lever-type door handle to replace a round knob, a remote control to operate a television, energy-efficient devices to replace inefficient ones*).

A2. Process Planning

By the end of this course, students will:

- A2.1** explain the need for a detailed production plan when manufacturing a product;
- A2.2** identify and analyse the factors that must be considered when planning production (*e.g., quantity, tolerances, cost factors*);
- A2.3** explain the differences in planning for custom-build versus mass-production processes;
- A2.4** apply collaborative problem solving to plan a detailed product manufacturing process (*e.g., including details of work flow, just-in-time part routing, scheduling, timelines, deadlines, efficient use of materials*).

A3. Control Systems

By the end of this course, students will:

- A3.1** describe various types of control systems (*e.g., programmable logic control [PLC], computer numerical control [CNC], inventory/purchasing control*);
- A3.2** incorporate the use of a control system (*e.g., Program Evaluation and Review Technique [PERT] chart, Gantt chart*) in a production process or project;
- A3.3** explain how a quality control system (*e.g., statistical process control, standards developed by the International Organization for Standardization [ISO]*) is used to manage production;

A3.4 utilize automation technology (*e.g., programmable logic controller [PLC], computer-aided manufacturing [CAM], pneumatic actuators*) to control a process.

A4. Mathematics, Science, and Technological Literacy

By the end of this course, students will:

A4.1 apply mathematical skills (*e.g., calculate production rates, machine speeds and feeds, ratios; use formulas related to geometry and trigonometry; estimate costs*) in planning manufacturing processes;

A4.2 apply scientific concepts to manufacturing-related processes (*e.g., material selection, material processing, setting of machining speed and cutting force*);

A4.3 apply technological literacy skills to create process control charts that clearly outline the stages of production;

A4.4 prepare engineering reports (*e.g., flow charts, inspection reports, tool lists, schedules, build plans*), using appropriate technical language to communicate the specifics of a product or process.

B. MANUFACTURING TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** identify and use appropriate processes to organize and control the manufacturing of products;
- B2.** demonstrate an understanding of the appropriate selection of materials to manufacture products to meet specific needs;
- B3.** demonstrate the safe and appropriate use of tools and equipment in the manufacture of a product or the development of a production process;
- B4.** analyse, properly operate, and maintain quality in a manufacturing system or process.

SPECIFIC EXPECTATIONS

B1. Managing the Manufacturing Process

By the end of this course, students will:

- B1.1** select the most efficient production method or process to manufacture a product;
- B1.2** develop managerial and mechanical control plans (*e.g., managerial: marketing, production tracking, financial control, inventory control, work-in-process tracking; mechanical control: electrical, pneumatic*) for the manufacture of a product (*e.g., robot, control system*);
- B1.3** implement managerial and mechanical control plans for the manufacture of a product;
- B1.4** use appropriate orthographic and isometric drawing techniques (*e.g., sketching, computer-aided design [CAD]*) when developing solutions to design problems or challenges.

B2. Material Selection

By the end of this course, students will:

- B2.1** demonstrate an understanding of destructive and non-destructive tests to evaluate material properties (*e.g., file test, spark test, moisture content test, dye testing, hardness testing*);
- B2.2** evaluate material properties (*e.g., heat resistance, mass, strength*) as they relate to specific project needs;
- B2.3** research and evaluate the properties of various ferrous and non-ferrous materials before

selecting the most appropriate material for a project;

- B2.4** demonstrate a working knowledge of the methods of conditioning materials (*e.g., thermal, chemical, mechanical methods*).

B3. Tools, Equipment, and Manufacturing Processes

By the end of this course, students will:

- B3.1** select and use the proper tools (*e.g., height gauge, surface gauge, trammel points, protractor, scribe*) and procedures required to effectively lay out a project;
- B3.2** demonstrate the proper use of various measuring instruments (*e.g., scale, vernier caliper, micrometer*), making measurements in both metric and US customary/British imperial units;
- B3.3** demonstrate the safe and proper use of appropriate hand tools, machine tools, and equipment (*e.g., files, wrenches, hack saws, electric drills, engine lathe, milling machine, planer, jointer, drill press, welder*) during a manufacturing process or in the assembly or fabrication of a product;
- B3.4** use a computer to operate and control a manufacturing system (*e.g., a robot, computer numerical control [CNC] equipment, a programmable logic control [PLC] device*) during a manufacturing process or in the fabrication of a project.

B4. Quality Assurance

By the end of this course, students will:

B4.1 analyse and explain production results in a manufacturing process (*e.g., by preparing a quality report*);

B4.2 use a quality control method (*e.g., statistical process control*) to monitor and record the reliability of a manufacturing process;

B4.3 explain how positive and negative attitudes influence production efficiency.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of ways in which the manufacturing industry affects the environment;
- C2.** demonstrate an understanding of ways in which the manufacturing industry affects society.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** analyse the effects that various manufacturing activities have on the environment (*e.g., the effects of waste disposal, power consumption, processing of raw materials; effects on Aboriginal hunting and gathering grounds*);
- C1.2** explain the benefits of using environmentally friendly processes and products in the manufacturing process;
- C1.3** explain how various sources of power generation (*e.g., coal, nuclear, wind, solar, geothermal*) and transportation methods (*e.g., truck, rail, ship*) used in manufacturing affect the environment.

C2. Technology and Society

By the end of this course, students will:

- C2.1** describe the social and economic consequences that a manufacturing activity can have or has had on individuals and society, including Aboriginal communities;
- C2.2** explain how the manufacturing industry affects the local and provincial economy (*e.g., with respect to job creation, standards of living, sustainability and conservation of the environment, impact on First Nation communities*);
- C2.3** describe recent trends in the local manufacturing industry (*e.g., globalization, rise in energy costs, increase in environmental awareness*) and their effect on the local community or the province as a whole, and predict future trends.

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and compliance with the health and safety legislation, standards, and practices that are essential to the safe operation of a manufacturing facility;
- D2.** demonstrate an understanding of postsecondary pathways that lead to career opportunities in the manufacturing industry.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** identify and explain the importance of the specific components of legislation and standards related to workplace safety in the manufacturing industry (e.g., *Occupational Health and Safety Act [OHSA]*, *Workplace Hazardous Materials Information System [WHMIS]*) and the specific responsibilities of the relevant oversight/regulatory organizations (e.g., *Workplace Safety and Insurance Board [WSIB]*, *Industrial Accident Prevention Association [IAPA]*);
- D1.2** demonstrate good housekeeping practices in the work environment (e.g., *cleaning up spills and leaks, keeping areas clean and clear of obstructions, properly organizing tools and equipment*);
- D1.3** handle materials safely and appropriately in compliance with the information included in the Material Safety Data Sheets (MSDS) from the Workplace Hazardous Materials Information System (WHMIS);
- D1.4** demonstrate an understanding of procedures to ensure safe and productive work practices in the manufacturing workplace (e.g., *perform safety inspections and audits that include ergonomic considerations related to workshop layout and set-up, material handling, ease of movement, lighting, workstation set-up*);
- D1.5** demonstrate the safe use of tools and equipment in compliance with safety manuals, instructions, and institutional requirements;

- D1.6** use proper ventilation and/or filtration systems to control air quality (e.g., *to minimize the effects of welding fumes, plastic off-gassing, cutting-fluid misting, heat treating of metal*);

- D1.7** use protective clothing and equipment as required to ensure their own and others' safety in the work environment.

D2. Career Opportunities

By the end of this course, students will:

- D2.1** identify employment opportunities available in the manufacturing industry locally, provincially, and nationally;
- D2.2** identify and describe the specific education and training requirements necessary for careers in the manufacturing industry (e.g., *technician, technologist, millwright, tool and die maker, mechanical engineer*);
- D2.3** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the manufacturing industry (e.g., *mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);
- D2.4** demonstrate an understanding of and apply the Essential Skills that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (e.g., *decision making, job task planning and organizing, problem solving*);

D2.5 demonstrate an understanding of and apply the work habits that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (*e.g., teamwork, reliability, initiative*);

D2.6 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in manufacturing technology (*e.g., Passport to Safety certificate, Ontario Skills Passport Work Plan, photographs of a project, technical drawings, reports, assignments*), and explain why having a current portfolio is important for career development and advancement.